

NATURE CURE

A project report submitted in partial fulfillment of the
Requirements for the award of degree of

Bachelor of Technology in **Computer Science and Engineering**

By

N190971-G.Sai Deekshitha
N190970-T.Sravanthi
N190067-K.V.Nagamani Devi
N190087-B.Lakshmi Prasanna
N190011-N.Sneha Madhuri
N190727-K.Lavanya

Under the Esteem Guidance of

Mrs.Y.Kalavathi



**DEPARTMENT OF COMPUTER
SCIENCE AND ENGINEERING**

**Rajiv Gandhi University of Knowledge Technologies –Nuzvid, Eluru,
Andhra Pradesh – 521202**



RAJIV GANDHI UNIVERSITY OF KNOWLEDGE TECHNOLOGIES
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
(A.P. Government Act 18 of 2008)
RGUKT-NUZVID, Eluru Dist. – 521202
Tele Fax: 08656 – 235557/235150

CERTIFICATE OF COMPLETION

This is to certify that the work entitled, “**NATURE CURE**” is the bonafied work of **G.Sai Deekshitha** with ID No:**N190971** and **T.Sravanthi** with ID No: **N190970** and **K.V.Nagamani Devi** with ID No: **N190067** and **B.Lakshmi Prasanna** with ID No: **N190087** and **N.Sneha Madhuri** with ID No: **N190011** and **K.Lavanya** with ID No: **N190727** carried out under my guidance and supervision for the partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in the department of Computer Science and Engineering in RGUKT Nuzvid. This work is done during the academic session July 2024 – November 2024, under my guidance.

Mrs. Y.Kalavathi

Project Supervisor
Faculty, Dept. of CSE
RGUKT , Nuzvid

Dr. D.V.Nagarjuna Devi

Head of CSE Department
Assistant Professor, Dept. of CSE
RGUKT , Nuzvid



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Tele Fax: 08656 – 235557/235150

CERTIFICATE OF EXAMINATION

This is to certify that the work entitled, “**NATURE CURE**” is the bonafied work of **G.Sai Deekshitha** with ID No: **N190971** and **T.Sravanthi** with ID No: **N190970** and **K.V.Nagamani Devi** with ID No: **N190067** and **B.Lakshmi Prasanna** with ID No: **N190087** and **N.Sneha Madhuri** with IDNo: **N190011** and **K.Lavanya** with ID No: **N190727** here by accord our approval of it as a study carried out and presented in a manner required for its acceptance in the partial fulfillment of the requirement for the award of the degree of Bachelor of Technology for which it has been submitted. This approval does not necessarily endorse or accept every statement made, opinion expressed or conclusion drawn, as a recorded in this thesis. It only signifies the acceptance of this thesis for the purpose for which it has been submitted.

Mrs.Y.Kalavathi

Project Supervisor

Faculty Dept. of CSE

RGUKT , Nuzvid

Dr. D.V.Nagarjuna Devi

Project Examiner

Faculty Dept. of CSE

RGUKT , Nuzvid



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Tele Fax: 08656 – 235557/235150

DECLARATION

We, **G.Sai Deekshitha** with ID No: **N190971** and **T.Sravanthi** with ID No: **N190970** and **K.V.Nagamani Devi** with ID No: **N190067** and **B.Lakshmi Prasanna** with ID No: **N190087** and **N.Sneha Madhuri** with ID No: **N190011** and **K.Lavanya** with ID No: **N190727** hereby declare that the project report entitle “**Nature Cure**” done by us under the guidance of **Mrs.Y.Kalavathi** is submitted for the partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering in the academic session July2024 – November 2024 at RGUKT – Nuzvid. We also declare that this project is a result of our own effort and has not been copied or imitated from any source. Citations from any websites are mentioned in the references. The results embodied in this project report have not been submitted to any other university or institute for the award of any degree or diploma.

Date :	G.Sai Deekshitha	(N190971)
Place: Nuzvid	T.Sravanthi	(N190970)
	K.V.Nagamani Devi	(N190067)
	B.LakshmiPrasanna	(N190087)
	N.Sneha Madhuri	(N190011)
	K.Lavanya	(N190727)

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G.Sai Deekshitha	(N190971)
T.Sravanthi	(N190970)
K.V.Nagamani Devi	(N190067)
B.Lakshmi Prasanna	(N190087)
N.Sneha Madhuri	(N190011)
K.Lavanya	(N190727)

ABSTRACT

This project presents a ResNet-18 based model to diagnose skin diseases from images and recommend relevant medicinal plant remedies. Recognizing the need for accessible diagnostic tools, especially in areas with limited dermatological expertise, we developed a system that classifies ten common skin disease types using a ResNet18 model trained from scratch on a comprehensive dataset.

Unlike approaches relying on transfer learning, our model is computationally efficient and straightforward. For each diagnosis, the system suggests natural plant-based remedies, making it a valuable resource for healthcare providers and individuals seeking alternative treatments. By combining advanced image classification with traditional medicine, this application offers an accessible, holistic solution for managing skin conditions.

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1.INTRODUCTION

This project is motivated by the high prevalence of skin diseases and the limited access to specialized dermatology care in many regions, which can delay diagnosis and treatment. Geographical and financial barriers often prevent timely healthcare, leading to prolonged symptoms.

Additionally, there is growing interest in accessible, natural treatment options, like medicinal plants, particularly in resource-limited areas. By creating an accurate, ResNet-18 based diagnostic tool that also suggests plant-based remedies, this project addresses the need for accessible healthcare and holistic treatment options, empowering both individuals and healthcare providers in managing skin conditions.

1.1 Problem Statement

Access to affordable and easy-to-use tools for diagnosing skin diseases is limited, especially in areas with few dermatologists. Many people are unable to get timely treatment, and current solutions are often too costly or complex. A system that can identify skin diseases and suggest natural remedies is needed to address this gap.

2. ANALYSIS OF EXISTING SYSTEM

-No existing system

3. PROPOSED METHODS

3.1. Dataset Collection

- Collected and organized a diseased skin image dataset into folders by disease classes.
- Applied preprocessing: resizing (224x224), normalization, and data augmentation.

3.2. Data Preprocessing

Data preprocessing is crucial for ensuring that the dataset is in a format suitable for training machine learning models.

- **Cleaning:** Remove duplicate and irrelevant entries.
- **Normalization:** Standardize the features (e.g., scale values like URL length, number of characters).
- **Handling Missing Data:** Fill or remove any missing data points.
- **Splitting Data:** Split the dataset into following datasets:
 - Training Set – 20370 samples – 75%
 - Testing Set - 3890 samples – 14%
 - Validation set – 2893 samples – 11%

3.3. Model Training in Kaggle

- Used ResNet18 for transfer learning with pre-trained weights.
- Replaced the top layer for 10-class classification.
- Compiled with Cross Entropy loss and Adam optimizer.
- Trained the model for 10 epochs, saved in **.pth** format.

3.4. Model Evaluation in Google Colab

- Loaded the trained ResNet1-8 model.
- Evaluated on test data.
- Tested predictions on individual sample images to verify correctness.

3.5. Flask Framework Development

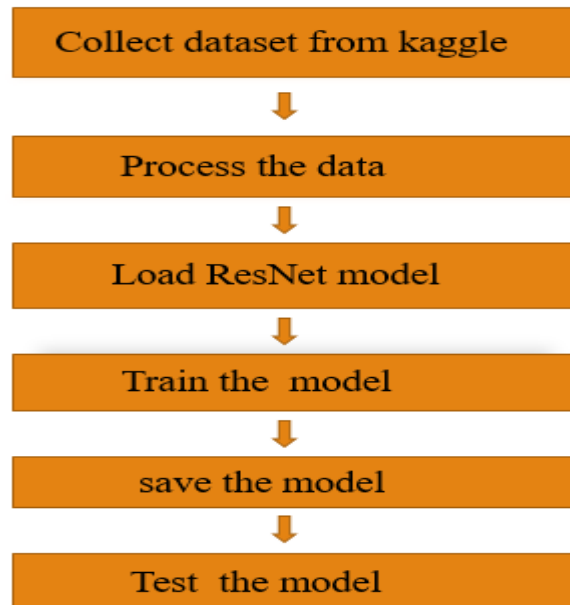
- Built a Flask-based web application: app.py
- Frontend: index.html (file upload), result.html (prediction output).
- Backend: Integrated the trained model to process input images.
- Predicted disease maps to a medicinal plant via a dictionary.
- Preprocessed uploaded images and displayed results dynamically.

3.6. Testing and Deployment

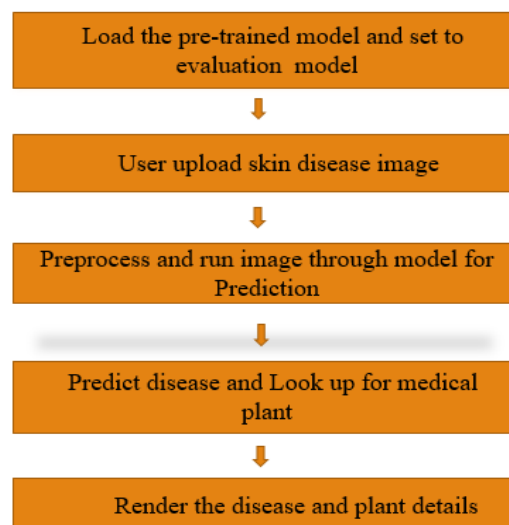
- Tested Flask app locally with diverse images for accuracy.
- Deployed the application.

4. FLOW CHART

END-TO-END WORKFLOW



FLASK BASED WORKFLOW



5. ALGORITHM USED

CNN Using ResNet for Image Classification

Type:

- **Supervised Learning Algorithm:** Trained on labeled data to classify images into predefined categories.
- **Convolutional Neural Network (CNN):** Specifically, a deep CNN architecture with residual blocks (ResNet).

Description:

ResNet (Residual Network) is a specialized CNN designed for deep learning tasks. It uses residual blocks to overcome the vanishing gradient problem, allowing the training of very deep networks. In image classification, ResNet processes images through layers of convolutions, pooling, and residual connections, extracting features to classify them into predefined categories.

- Pre-trained ResNet models like **ResNet-18**, **ResNet-34**, or **ResNet-50** are often fine-tuned on a specific dataset for classification tasks.

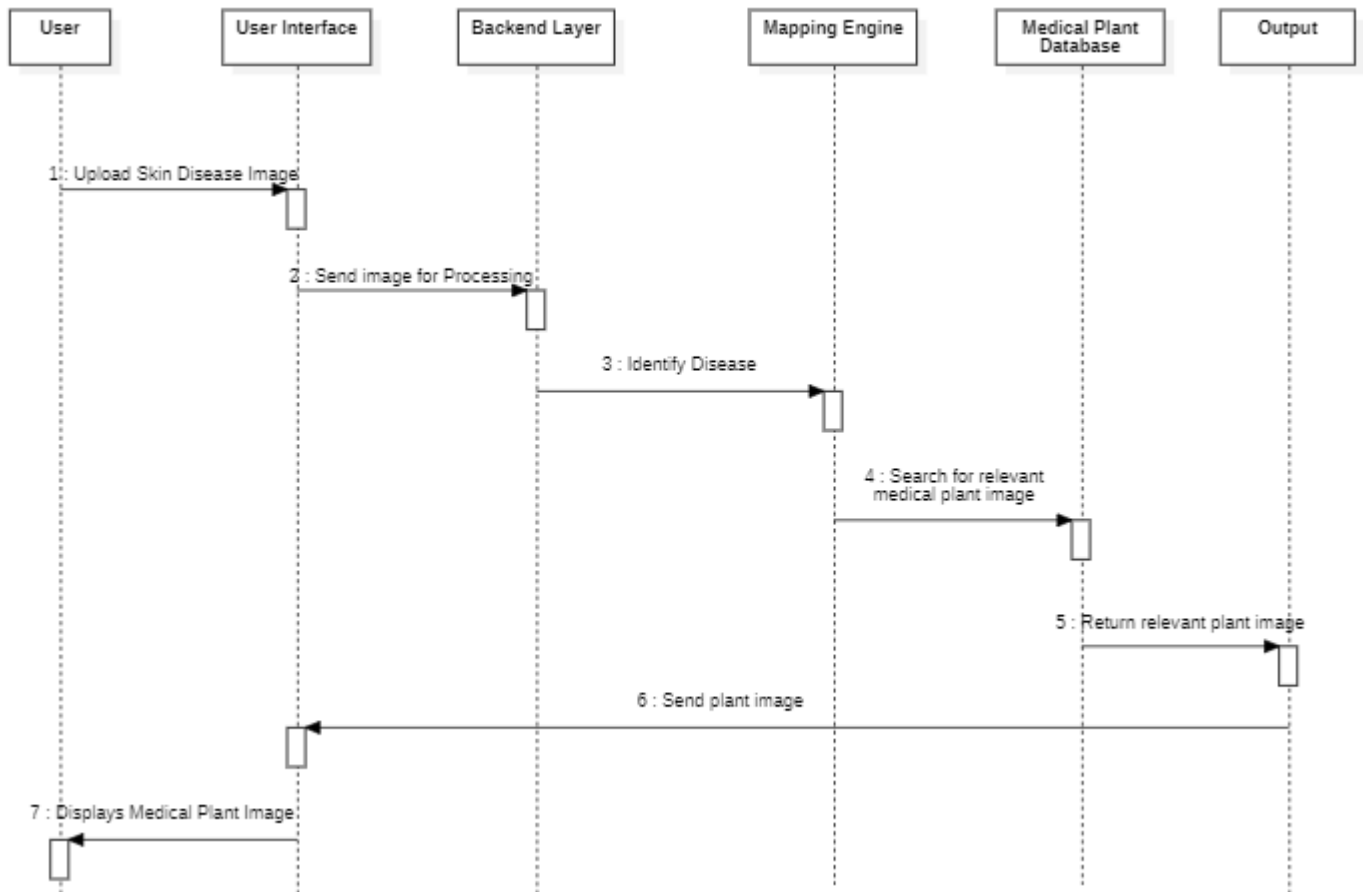
Advantages:

1. **Handles Deep Networks Effectively:** Residual connections solve vanishing gradient issues.
2. **Pre-trained Weights:** Easily fine-tuned for transfer learning, saving time and computation.
3. **High Accuracy:** Delivers superior performance for image classification tasks.
4. **Scalability:** Supports very deep architectures, improving feature extraction.

Disadvantages:

1. **High Computational Cost:** Deep architecture requires significant resources.
2. **Overfitting Risk:** Fine-tuning on small datasets may lead to overfitting.
3. **Complexity:** Network design and optimization can be challenging for new users.
4. **Dependency on Large Datasets:** Pre-training often requires large datasets like ImageNet.

6. ARCHITECTURE



7. REQUIREMENTS

7.1. Software Components

Frontend:

- **Framework:**
 - HTML,CSS, and Bootstrap for basic user Interface.

Backend:

- **Framework:**
 - Python with Flask for handling request and ML running models
- **Flask:** Flask is a micro web framework for Python that provides the essential tools and features to build web applications, APIs, and dynamic websites. It is designed to be simple, extensible, and easy to use, offering a minimalistic core while allowing developers to add functionality through a variety of extensions and libraries.

DataBase:

- Kaggle Dataset for skin disease

Machine Learning:

- **Models Used:**
 - Resnet-18.
- **Library Support:**
 - Torch
 - NumPy
 - Pandas
 - Seaborn
 - Matplotlib
- **Model Artifacts:**
 - Saved model file(resnet.pth) for deployment

7.2. Hardware Components

Flask itself is lightweight and does not impose heavy hardware requirements. The hardware needed primarily depends on the expected workload and traffic your Flask application will handle.

- **CPU:** A modern multi-core processor is recommended for handling concurrent requests efficiently.
- **RAM:** Flask applications typically do not require significant memory unless you're reprocessing large amounts of data or running complex operations. Generally, a minimum of 2GB RAM is sufficient for development and small-scale deployments.
- **Storage:** Flask applications do not consume much storage space by themselves. Ensure you have adequate disk space for storing application code, log files, and any data files your application might generate or utilize.
- **Keyboard:** A Standard keyboard for user input

8. CONCLUSION

8.1 Result

"Nature Cure" project has shown promising results, with a training accuracy of 91% and a test accuracy of 64%. However, there is still room for improvement in the model's ability to generalize to unseen data. Additionally, the integration of a Flask-based web framework offers a user-friendly interface, making it accessible for practical deployment. With these improvements, the project has the potential to evolve into a robust and reliable system for skin disease classification and medicinal plant recommendations.

8.2 Future Enhancement

To further enhance the accuracy, we plan to expand and diversify the dataset, implement advanced data augmentation techniques, and fine-tune the model's parameters. Additionally, we aim to refine the training process and explore other strategies to improve performance. The model has been successfully integrated into a Flask-based web framework, providing users with medicinal plant recommendations based on skin images. the focus will be on optimizing the model to achieve higher accuracy and better practical outcomes.

9. REFERENCES

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